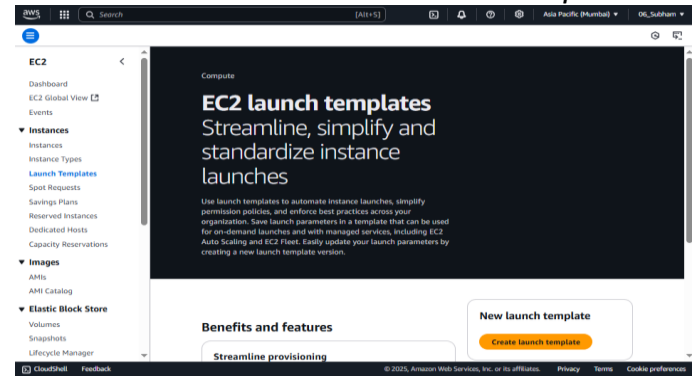


ASSIGNMENT 11: Build scaling plans in AWS that balance the load on different EC2 instances.

Step 1: Login to your AWS account and go to **'Launch Templates'** under **Instances of EC2**. Then click on **'Create launch templates'**.



Step 2: Fill up the details as follows and click on **'Create launch templates'**.

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - required

ti-off

Must be unique to this account. Max 128 chars. No spaces or special characters like '!', '"', '\', or ' '.

Template version description

v0

Max 255 chars

Auto Scaling guidance | info

Select this if you intend to use this template with EC2 Auto Scaling

☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

Template tags

Source template

Launch template contents

Specify the details of your launch template below. Leaving a blank field will result in the field not being included in the launch template.

Application and OS Images (Amazon Machine Image) - required | info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recent | **Quick Start**

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian | **Browse more AMIs**

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), 55D Volume Type
ami-0e35dab05955cf57 (64-bit (x86)) / ami-042b6b610d41c0a0 (64-bit (ARM))
Virtualization: hvm | Disk: enabled: true | Root device type: ebs

Description

Ubuntu Server 24.04 LTS (HVM), EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture | **AMI ID** | **Publish Date** | **Username** | **Verified provider**

64-bit (x86) | ami-0e35dab05955cf57 | 2025-03-05 | ubuntu | **Verified provider**

Instance type | **Get advice** | **Advanced**

Instance type

t2.micro | Family: t2 | 1 vCPU | 1 GB Memory | Current generation: true
On-Demand Linux base pricing: 0.0124 USD per Hour
On-Demand Windows base pricing: 0.017 USD per Hour
On-Demand Linux base pricing: 0.0064 USD per Hour
On-Demand Windows base pricing: 0.0142 USD per Hour
On-Demand Linux base pricing: 0.0114 USD per Hour

Additional costs apply for AMIs with pre-installed software

Key pair (login) | info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

keyG7 | **Create new key pair**

Network settings | info

Subnet | info

Don't include in launch template | **Create new subnet**

When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups) | info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Select existing security group | ☒ **Create security group**

Security group name - required

7G

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, 0-9, hyphen, and underscore. /0a-g/0+==&015

Description - required | info

SG_7g

VPC | info

vpc-02493897f3c47f3e0 (default) | **Create new VPC**

Inbound Security Group Rules

Security group rule 1 (TCP, 22, 0.0.0.0/0) | **Remove**

Type | info | **Protocol** | info | **Port range** | info

Custom TCP | TCP | 0

Source type | info | **Source** | info | **Description - optional** | info

Custom | 0.0.0.0/0 | e.g. SSH for admin desktop

Security group rule 2 (TCP, 80, 0.0.0.0/0) | **Remove**

Type | info | **Protocol** | info | **Port range** | info

HTTP | TCP | 80

Source type | info | **Source** | info | **Description - optional** | info

Custom | 0.0.0.0/0 | e.g. SSH for admin desktop

Security group rule 3 (TCP, 443, 0.0.0.0/0) | **Remove**

Type | info | **Protocol** | info | **Port range** | info

HTTPS | TCP | 443

Source type | info | **Source** | info | **Description - optional** | info

Custom | 0.0.0.0/0 | e.g. SSH for admin desktop

Security group rule 4 (TCP, 22, 0.0.0.0/0) | **Remove**

Type | info | **Protocol** | info | **Port range** | info

SSH | TCP | 22

Source type | info | **Source** | info | **Description - optional** | info

Custom | 0.0.0.0/0 | e.g. SSH for admin desktop

Add security group rule

Advanced network configuration

Storage (volumes) | info

EBS Volumes | **Hide details**

Volume 1 (AMI Root) - 8 GiB, EBS, General purpose SSD (gp3)
AMI Volumes are not included in the template unless modified

Add new volume

The selected AMI contains more instance store volumes than the instance allows. Only the first 0 instance store volumes from the AMI will be accessible from the instance

Resource tags | info

No resource tags are currently included in this template. Add a resource tag to include it in the launch template.

Add new tag

You can add up to 50 more tags.

Advanced details | info

IAM instance profile | info

Don't include in launch template | **Create new IAM profile**

Hostname type | info

Don't include in launch template

DNS Hostname | info

☐ Enable resource-based IPv4 (A record) DNS requests
☐ Enable resource-based IPv6 (AAAA record) DNS requests

Instance auto-recovery | info

Don't include in launch template

Shutdown behavior | info

Don't include in launch template
Not applicable for EC2 Auto Scaling

Stop - Hibernate behavior | info

Don't include in launch template
Not applicable for Amazon EC2 Auto Scaling

Termination protection | info

Don't include in launch template

Spot protection | info

Don't include in launch template

Detailed CloudWatch monitoring | info

Don't include in launch template

Credit specification | info

Don't include in launch template

Placement group | info

Don't include in launch template | **Create new placement group**

EBS-optimized instance | info

Don't include in launch template

Instance bandwidth configuration | info

Don't include in launch template

Purchasing option | info

☒ Don't include in launch template
☐ Capacity Blocks
Launch instances for your active capacity blocks
☐ Spot instances
Request Spot Instances at the Spot price, capped at the On-Demand price
If Spot is selected you will not be able to create an Auto Scaling group that spans across multiple pricing options and instance types

Capacity reservation | info

Don't include in launch template

Tenancy | info

Don't include in launch template

RAM disk ID | info

Don't include in launch template

Kernel ID | info

Don't include in launch template

Nitro Enclave | info

Don't include in launch template
Nitro Enclaves are not compatible with instance types that have less than 2 vCPUs.

License configurations | info

Don't include in launch template

CPU options | info

Configure CPUs for your instance to optimize performance and save on licensing costs.
☐ Use default CPU options
☐ Specify CPU options
The t2.micro instance type does not support configuring CPUs. To view instance types that support CPU options, see [Supported CPU options](#) in the EC2 User Guide.

Default active vCPUs | **Total vCPUs**

1 | 1

Metadata accessible | info

Enabled

Metadata IPv6 endpoint | info

Don't include in launch template

Metadata version | info

V2 only (token required)

For V2 requests, you must include a session token in all instance metadata requests. Applications or agents that use V1 for instance metadata access will break.

Metadata response hop limit | info

2

Allow tags in metadata | info

Don't include in launch template

User data - optional | info

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
apt-get update -y
apt-get upgrade -y
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
apt-get install -y git
curl -sL https://deb.nodesource.com/setup_22.x | sudo -E bash -
apt-get install -y nodejs
git clone https://github.com/MirrorOfMam-MS427/DS.git
cd DS
npm install
node index.js
```

☐ User data has already been base64 encoded

Summary

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64, read more
ami-0e35dab05955cf57

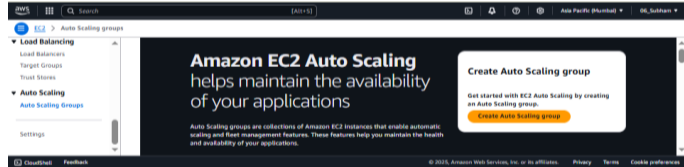
Virtual server type (Instance type)
t2.micro

Firewall (security group)
New security group

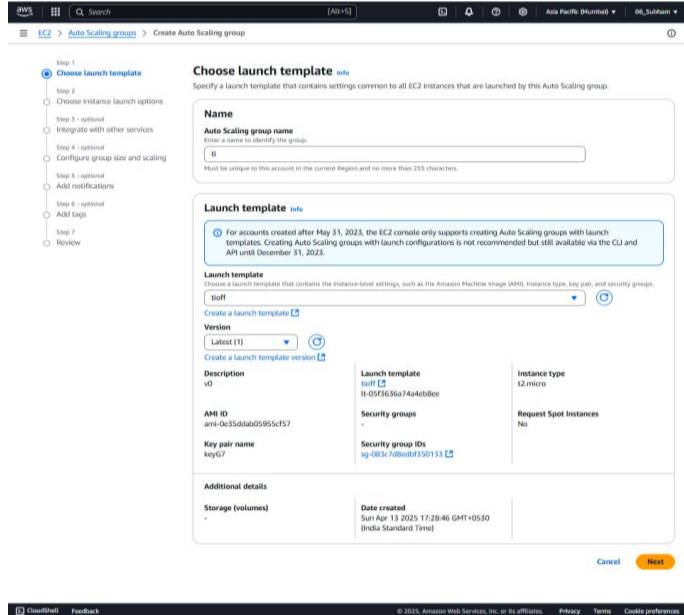
Storage (volumes)
1 volume(s) - 8 GiB

Create launch template

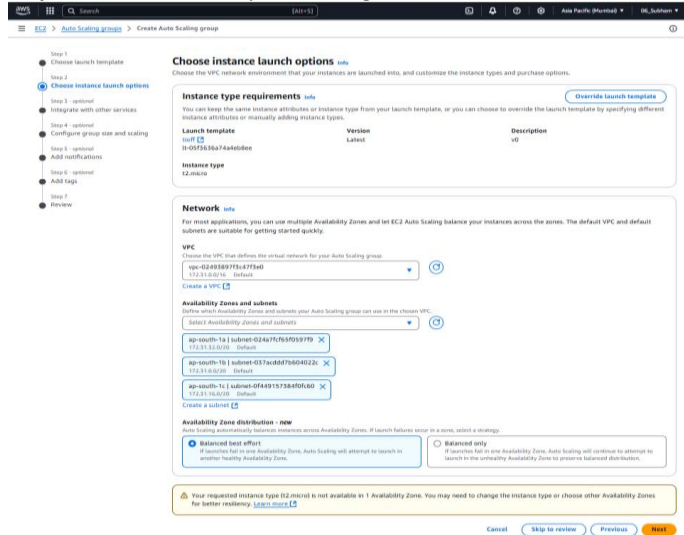
Step 3: Now go to on 'Auto Scaling Groups' under Auto Scaling. Then Click on 'Create Auto Scaling group'.



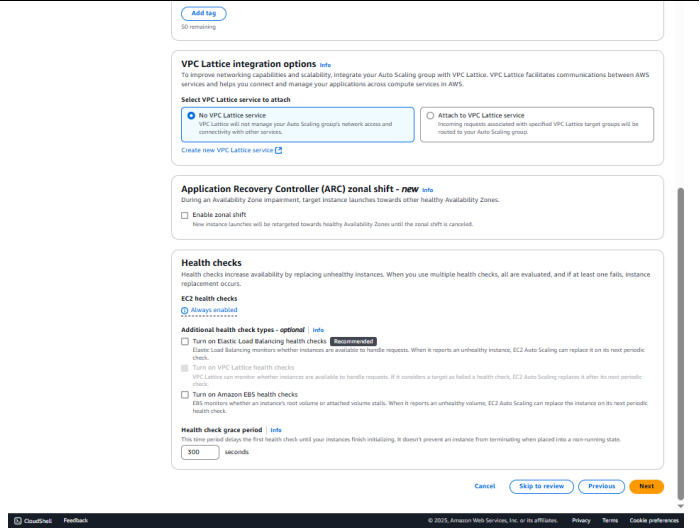
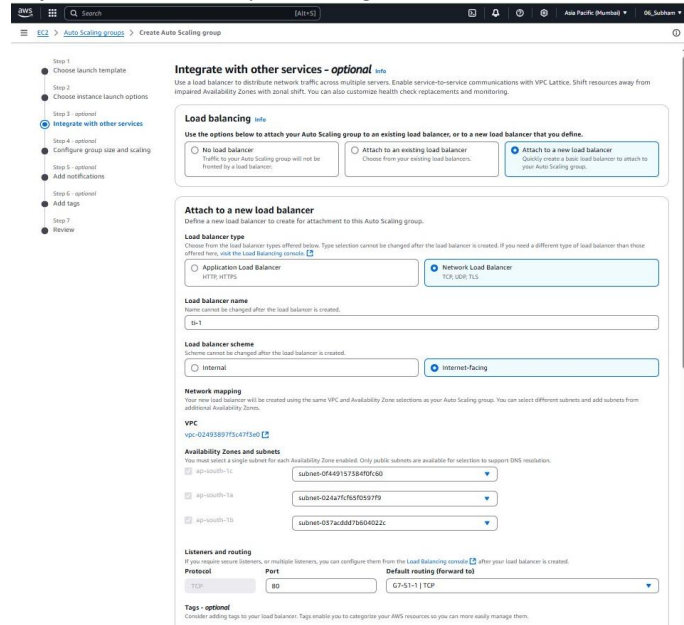
Step 4: Fill the details as per the image and click on 'Next'.



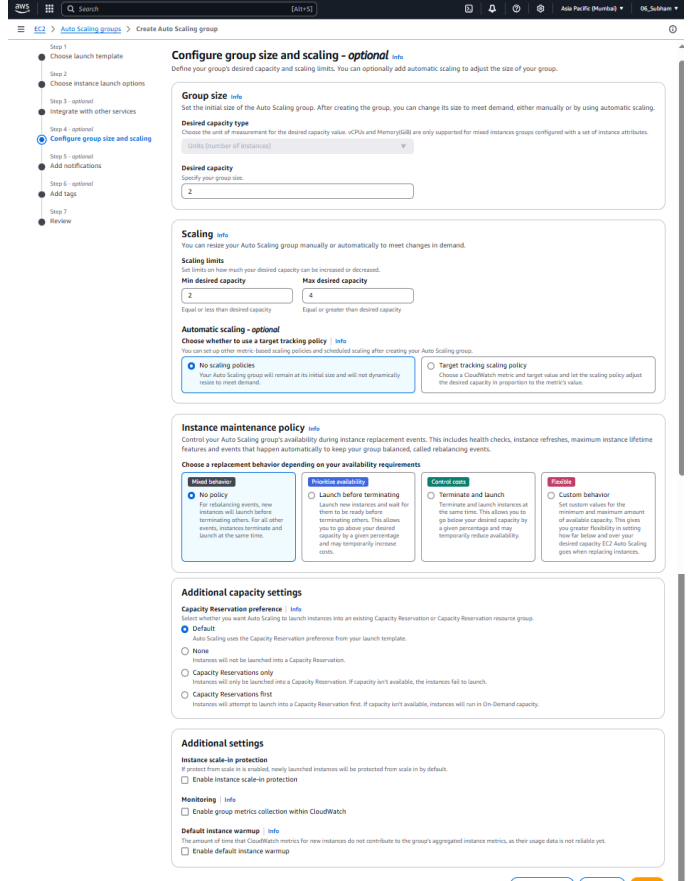
Step 5: Fill the details as per the image and click on 'Next'.



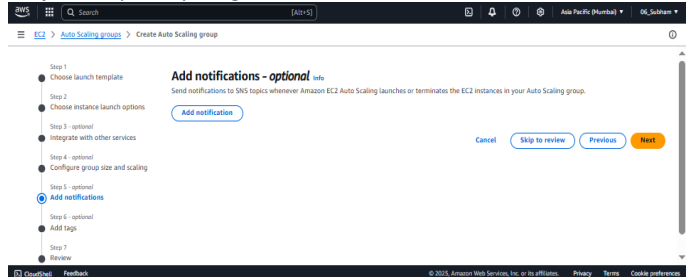
Step 6: Fill the details as per the image and click on 'Next'.



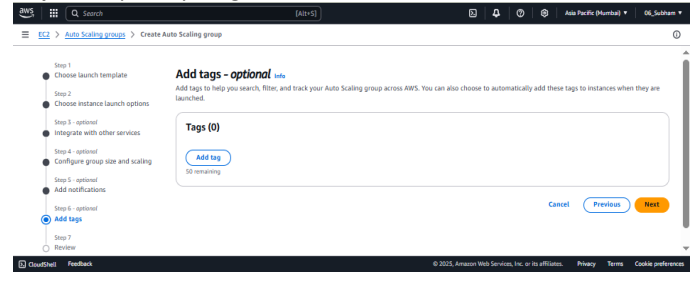
Step 7: Fill the details as per the image and click on 'Next'.



Step 8: Keep everything as it is and click on 'Next'.



Step 9: Keep everything as it is and click on 'Next'.



Step 10: Now check the Review; click on 'Create Auto Scaling group'.

The screenshot shows the 'Review' step of the 'Create Auto Scaling group' wizard. It includes sections for 'Step 1: Choose launch template', 'Step 2: Choose instance launch options' (showing network details like VPC, subnets, and availability zones), 'Step 3: Integrate with other services' (showing load balancing, VPC lattice, and ARC), and 'Health checks'.

The screenshot shows 'Step 4: Configure group size and scaling policies'. It includes settings for 'Group size' (desired capacity), 'Scaling' (minimum and maximum desired capacity, target tracking policy), 'Instance maintenance policy' (replacement behavior, min/max healthy percentages), 'Additional settings' (instance scale-in protection, monitoring, default instance warmup), 'Capacity Reservation preference' (preference, capacity reservation IDs, resource groups), 'Step 5: Add notifications', and 'Step 6: Add tags'.

Step 11: After creating Auto Scaling group, Instances create automatically.

The screenshot shows the 'Instances' page in the AWS Management Console. It displays a list of two EC2 instances in a 'Running' state, with details like instance ID, name, state, type, and availability zone.

Step 12.a: Click an instance; copy IPv4 address from Instance details.

The screenshot shows the 'Instance details' page for instance `i-00e9ba6d3fa7b8d0`. It displays various tabs like Details, Status and alarms, Monitoring, Security, Networking, Storage, and Tags. The 'Details' tab is active, showing instance ID, name, state, type, and public/private IP addresses.

Step 12.b: Now using the IP Address login to the Bitwise SSH Client and Open the Terminal give the command 'nano infil.sh'.

```
ubuntu@3.109.139.251:22 - Bitwise xterm - ubuntu@ip-172-31-33-247: ~
Last login: Sun Apr 13 12:24:04 2025 from 49.37.34.2
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-33-247:~$ nano infil.sh
```

Step 12.c: After opening nano editor type the code as given; then press 'Ctrl+O' & hit enter to save and then press 'Ctrl+X' to exit.

```
GNU nano 7.2 infil.sh
#!/bin/bash
while true
do
    echo "LOADING"
done
```

Step 12.d: Run the command 'chmod +x infil.sh' and './infil.sh' respectively.

```
ubuntu@ip-172-31-33-247:~$ nano infil.sh
ubuntu@ip-172-31-33-247:~$ chmod +x infil.sh
ubuntu@ip-172-31-33-247:~$ ./infil.sh
```

Step 12.e: The file runs into an infinite loop.

```
ubuntu@3.109.139.251:22 - Bitwise xterm - ubuntu@ip-172-31-33-247: ~
LOADING
LOADING
LOADING
LOADING
LOADING
LOADING
```

Step 13.a: Click another instance; copy IPv4 address from the details.

The screenshot shows the 'Instance details' page for instance `i-039b70a78689b487e`. It displays various tabs like Details, Status and alarms, Monitoring, Security, Networking, Storage, and Tags. The 'Details' tab is active, showing instance ID, name, state, type, and public/private IP addresses.

Step 13.b: Now using the IP Address login to the Bitwise SSH Client and Open the Terminal give the command 'nano infill.sh'.

```
ubuntu@15.207.19.231:22 - Bitwise xterm - ubuntu@ip-172-31-8-168: ~
Last login: Sun Apr 13 12:31:54 2025 from 49.37.34.2
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-8-168:~$ nano infill.sh
```

Step 13.c: After opening nano editor type the code as given; then press 'Ctrl+O' & hit enter to save and then press 'Ctrl+X' to exit.

```
GNU nano 7.2 infill.sh
#!/bin/bash
while true
do
    echo "LOADING"
done
```

Step 13.d: Run the command 'chmod +x infill.sh' and './infill.sh' respectively.

```
ubuntu@ip-172-31-8-168:~$ nano infill.sh
ubuntu@ip-172-31-8-168:~$ chmod +x infill.sh
ubuntu@ip-172-31-8-168:~$ ./infill.sh
```

Step 13.e: The file runs into an infinite loop.

```
ubuntu@15.207.19.231:22 - Bitwise xterm - ubuntu@ip-172-31-8-168: ~
LOADING
LOADING
LOADING
LOADING
LOADING
LOADING
```

Step 14: Now select two instances and go to the Monitoring section, some graphs will appear.

